

1.

$$\begin{aligned} & \text{(a)} \\ & \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y} - \sum_{i=1}^n \bar{x} y_i + \sum_{i=1}^n \bar{x} \bar{y} \\ &= \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n y_i + \bar{x} \bar{y} \sum_{i=1}^n 1 \\ &= \sum_{i=1}^n x_i y_i - \bar{y}(n\bar{x}) - \bar{x}(n\bar{y}) + \bar{x}\bar{y}(n) \\ &= \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} \end{aligned}$$

$$\begin{aligned} & \text{(b)} \\ & \sum_{i=1}^n (x_i - \bar{x}) \\ &= \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} \\ &= \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n 1 \\ &= n\bar{x} - \bar{x}(n) = 0 \end{aligned}$$

Note: Because of the fact (b), we know that

$$\begin{aligned} & \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^n (x_i - \bar{x})(y_i) \end{aligned}$$

2. The objective function is $\min \sum_{i=1}^n \hat{u}_i^2$; here we don't need to find $\hat{\beta}_0$ as the model does not contain the intercept.

$$\min \sum_{i=1}^n \hat{u}_i^2 = \min \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \min \sum_{i=1}^n (y_i - \hat{\beta}_1 x_i)^2$$

Do note that we take the first derivative with respect to $\hat{\beta}_1$, NOT β_1 .

$$\frac{d \sum_{i=1}^n \hat{u}_i^2}{d \hat{\beta}_1} = 2 \times \sum_{i=1}^n (y_i - \hat{\beta}_1 x_i) \times (-x_i) = 0$$

$$\Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

3.

(a) For consumption and disposable income, it is a log-level case. So, we explain

$\hat{\beta}_{disposable_income}$ as follows.

*A one (unit of X) of X increase is associated with $(\hat{\beta}_1 * 100\%)$ increase/decrease in Y on average, ceteris paribus.*

A one (thousand) of income is associated with $(0.15*100\%)$ increase in consumption, ceteris paribus

(b) For consumption and interest rate, it is also a log-level case.

*A one (unit of X) of X increase is associated with $(\hat{\beta}_2 * 100\%)$ increase/decrease in Y on average, ceteris paribus.*

A one % of interest rate increase is associated with $(.065 * 100\%)$ decrease in consumption on average, ceteris paribus.

4.

(a)

Degree of freedom is $200-1-1=198$

The confidence interval formula is as follows.

$$\left[\hat{\beta}_1 \pm t_{(n-k-1), 1-\frac{\alpha}{2}} \times \text{se}(\hat{\beta}_1) \right] = \left[\hat{\beta}_1 \pm t_{198, 97.5\%} \times \text{se}(\hat{\beta}_1) \right]$$

Therefore, 95% Confidence Interval: $[2.5 \pm 1.96 \times 0.3]$

(b)

$$\left[\hat{\beta}_1 \pm t_{(n-k-1), 1-\frac{\alpha}{2}} \times \text{se}(\hat{\beta}_1) \right] = \left[\hat{\beta}_1 \pm t_{198, 99.5\%} \times \text{se}(\hat{\beta}_1) \right]$$

Therefore, 99% Confidence Interval: $[2.5 \pm 2.576 \times 0.3] = [1.727, 3.278]$

(c)

We are testing $H_0: \beta_1 = 2$ against $H_1: \beta_1 \neq 2$

The 95% Confidence Interval: $[2.5 \pm 1.96 \times 0.3] = [1.912, 3.088]$.

There is 95% of the chance this confidence interval contains the true value β_1 . Since 2 lies in the confidence interval, we cannot reject the null hypothesis that β_1 is 2.